⇒ BN reduces effects of exploding & vanishing gradients because Input to every layer becomes roughly normal distributed. W/o BN, low activations of one layer can lead to lower activations in the next layer & then even lower ones in the next layer & so on.

⇒ Normally, large learning rates may increase the scale of layer parameters, which then amplify the gradient during backpropagat & lead to the model explosion. With BN, backprop through a layer is unaffected by the scale of its parameters. BN stablize the parameter growth.

↳ We introduce, for activat^n $x^{(k)}$, a pair of parameters, $\gamma^{(k)}, \beta^{(k)}$ which scale & shift the normalized value:

$$y^{(k)} = \gamma^{(k)} \hat{x}^{(k)} + \beta^{(k)}$$

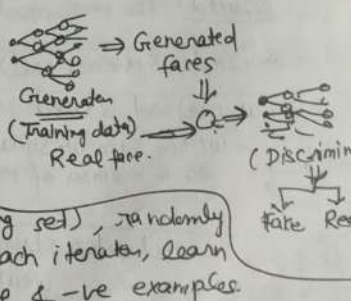
scaling & shifting parameters.

BN ⇒ FC → BN → tanh → FC → BN → tanh (BAD) Drop out
sequence (usually) [FC = Fully Connected layer] (Convolution layer) BN Activat

(A) GANs (Generative Adversarial N/ws)

Generative Deep Inferencing models = Generate a new object that closely matches to those in a limited training set by closely approximating the underlying (but unknown) prob. dist^n of set of objects.

→ Ex: GAN has 2 NN < Generator, Discriminator.



→ Generator's training depends on Discriminator
& Discriminator's " " Generator.

→ Discriminator Training

Given a set of +ve examples (good images from training set), randomly generate a set of −ve " (bad images) & In each iteration, learn a discriminator funct^n D that discriminate +ve & −ve examples.

⇒ Generator N/w : Try to fool the discriminator by generating real looking images
Discriminator N/w : try to distinguish b/w real & fake images

→ Train Jointly in minimax game, Discriminator wants to maximize objective funct^n to discriminate +ve & −ve data & Generator wants to minimize objective funct^n to less fool the (discrim

→ Use cross-entropy with any loss like Hinge Loss).

→ Use KL Divergence ⇒ measures how one prob. dist^n p diverges from a second expected prob. dist^n q.

$D_{KL}(P||Q) = \int_x P(x) \log \frac{P(x)}{Q(x)} dx$

⇒ BN reduces effects of exploding & vanishing gradients because Input to every layer becomes roughly normal distributed. w/o BN, low activations of one layer can lead to lower activations in the next layer & then even lower ones in the next layer & so on.

⇒ Normally, large learning rates may increase the scale of layer parameters, which then amplify the gradient during backpropagat^n & lead to the model explosion. With BN, backprop through a layer is unaffected by the scale of its parameters. BN stablize the parameter growth.

↳ We introduce, for eactivat^n $x^{(k)}$, a pair of parameters, $\gamma^{(k)}, \beta^{(k)}$ which scale & shift the normalized value:

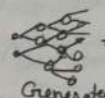
$$y^{(k)} = \gamma^{(k)} \hat{x}^{(k)} + \beta^{(k)}$$

scaling & shifting parameters.

BN ⇒ FC → BN → tanh → FC → BN → tanh (BAPD) Dropout, BN, Activat^n
sequence (usually) [FC = Fully Connected layer] (Convolution layer)

(*) GANs (Generative Adversarial N/ws)

Generative Deep Inferencing models = Generate a new object that closely matches to those in a limited training set by closely approximating the underlying (but unknown) prob. dist^n of set of objects.

→ GAN has 2 NN < Generator, Discriminator. ⇒ Random Noise →  ⇒ Generated faces

Generator (Training data) → Real face. ⇓ → (Discriminator) → Fake, Real.

→ Generator's training depends on Discriminator
& Discriminator's " " Generator

→ Discriminator Training
Given a set of +ve examples (good images from training set), randomly generate a set of −ve " (bad images) & In each iteration, learn a discriminator funct^n D that discriminate +ve & −ve examples.

⇒ Generator N/w : Try to fool the discriminator by generating real looking images
Discriminator N/w : try to distinguish b/w real & fake images

→ Train jointly in minimax game, Discriminator wants to maximize objective funct^n to discriminate +ve & −ve data & Generator wants to minimize objective funct^n to less fool the discrim.

→ Use cross-entropy with any loss like Hinge Loss). → $D_{KL}(P||Q) = \int_x p(x) \log \frac{p(x)}{q(x)} dx$

→ Use KL Divergence ⇒ measures how one prob. dist^n p diverges from a second expected prob. dist^n q.

(*) $D_{KL}(p \| q) = \int_x p(x) \log \frac{p(x)}{q(x)} dx$.

[D_{KL} achieves minimum zero when $p(x) == q(x)$ everywhere)

(*) Jensen-Shannon Divergence is another measure of similarity b/w 2 prob. dist's bounded by [0,1]. It is symmetric & more smooth.

$D_{JS}(p \| q) = \frac{1}{2} D_{KL}\left(p \| \frac{p+q}{2}\right) + \frac{1}{2} D_{KL}\left(q \| \frac{p+q}{2}\right)$

(*) Divergence is like distance measures.. KL Divergence is asymmetric

(*) Overfitting can be avoided by - taking simple model, fewer variables, remove noise in training data, use CV techniques such as K-fold CV., use regularizat^n techniques such as LASSO that penalize certain model parameters if they are likely to cause overfitting.

(*) Feature Selection Methods → Filter Methods (LDA, ANOVA, Chi-Square)
→ Wrapper Methods (Forward/Backward Selection, Recursive Feature Elimination)

(*) Maintenance of deployed Model.

① Monitor: Constant monitoring of all models to determine performance accuracy of model

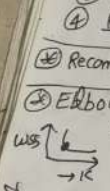
② Evaluate: Evaluat^n of metrics of the current model is calculated to determine if new algo^m is needed.

③ Compare: The new models are compared each other to find best one

④ Rebuild: The best model is rebuilt on the current state of data.

(*) Recommender Systems → ① Collaborative Filtering ② Content based Filtering

(*) Elbow method is used to select "K" in K-mean.



→ within sum of squares (WSS) is the sum of squared distance b/w each member of the cluster & its centroid. & K = no. of clusters.

(*) $p \leq 0.05$ → Indicates strong evidence against Null Hyp. & so reject Null Hyp.

$p > 0.05$ → " weak " " " So, fail to reject " "

$p = 0.05$ → Considered to be marginal (Could go either way).

(*) Random Forest is less affected by outlier. & it is also used for dimensionality reduct^n

(*) Exploding gradients It is a prob^m where large error gradients accumulate & result in a very large update of weights during training.

(*) Selection Bias occurs
representative of populat^n
introduced by select^n of
in such a way that proper

(*) In RF regression, instead

(*) A Box-Cox transform i
dependent variables into

(*) Regularizat^n is the p
model to induce smooth

(*) Data Wrangling (Process
↳ Steps. ① Data Profiling
② " Visualiz
③ Standarizat/N

(*) L2 (ridge) regularizat^n is less
always one soln. L1 i
can possibly have mul

(*) MSE gives a relativel
(mean squared error) MSE tends
A more robust alternat

(*) RF don't need cross

(*) Mean imputat^n is genera
account feature correlat^n
of the data & increases
accurate model & a na

(*) If data follows norma

(*) An inlier is a data
is unusual or an error &

(*) Randomizat^n controls

A confounding variable is
spurious association wher

(*) Lift. Measure of Performan

(*) Selection Bias occurs when sample obtained is not representative of population intended to be analysed. It is a bias introduced by selectn of individuals, groups or data for analysis in such a way that proper randomizatn is not achieved.

(*) In RF regression, instead of majority vote, we take mean/average

(*) A Box-Cox transform is a way to transform non-normal dependent variables into a normal shape.

(*) Regularizatn is the process of adding tuning parameter to a model to induce smoothness in order to prevent overfitting.

(*) Data Wrangling (Process of gathering, selecting & transforming).
↳ Steps. ① Data Profiling (find shape & description of data)
② " Visualizn (use histogram / boxplots & scatter plots)
③ Standarizat/Normal.- ④ Handling missing values, remove duplicates / irrelevant feature.

(*) L2 regularizatn (ridge) is less robust but has a stable solution & always one soln. L1 is more robust but has unstable soln & can possibly have multiple soln.

(*) MSE (mean square error) gives a relatively high weight to large errors — therefore, MSE tends to put too much emphasis on large deviations. A more robust alternative is MAE (Mean absolute deviation).

(*) RF don't need cross validatn | Power = P(reject null hyp | alternate hyp is true)

(*) Mean imputatn is generally a bad practice because it does not take into account feature correlatn. & also, mean imputatn reduces the variance of the data & increases bias in our data. This leads to a less accurate model & a narrower confidence interval due to a smaller variance.

(*) If data follows normal distn then using z-score, we find outliers.

(*) An inlier is a data observatn that lies within rest of the data set & is unusual or an error & it is harder to identify since it is in data.

(*) Randomizatn controls bias.

(*) A confounding variable is a variable that influences both dependent & independen variable & causing a spurious association where variables are not causally related.

(*) Lift. Measure of performance of target model against a random choice model.

ISLR

→ Suppose we are statistical consultant hired by a client to investigate the association b/w advertising budget/~~sales~~ and sales of a prod.

→ Sales = f (TV) or sales = f (Radio) or sales = f (Newspaper).

→ Client can control the advertising expenditure by finding best fitting

→ Our goal is to develop an accurate model that can be used to predict sales on these 3 media budgets.

→ Prediction/independent variables/features/variables = Inputs.

→ response/dependent variable = output.

→ Suppose Y = Quantitative Response & Prediction $X = (X_1, X_2, \ldots, X_p)$

then relationship can be written as $Y = f(X) + \varepsilon$ [f = fixed unknown fn

(ε = random error term which is independent of X & mean(ε) = 0]

→ other ex. ⇒ Income = f (Years of Educat^n, ~~Experience~~)

⇒ Statistical learning refers to a set of approaches estimating f.

⇒ There are 2 reasons for estimating f = Predict^n & Inference.

⇒ $\hat{Y} = \hat{f}(X)$ [$\hat{f}$ = estimate of f]

⇒ Ex. $X_1, X_2, \ldots, X_p$ are characteristics of a patient's blood sample which predicts important related Y is a variable encoding patient's risk for a severe adverse react^n to a drug

⇒ $\hat{f}$ will not be a perfect estimate for f ⇒ induced reducible error.

⇒ Since $Y = f(x) + \varepsilon$, so, variability associated with ε also affects the predict^n

It is called irreducible error. Ex. risk of adverse react^n depends on manufacturing variation in drug or patient's general feeling on that day

⇒ n diff. data points (Training data) is used to train/teach our method to estimate f

⇒ Parametric Methods – We make assumpt^n about functional form or shape of f. (eg. Linear model) only estimate coeff. $\hat{\beta}_0, \hat{\beta}_1, \ldots$

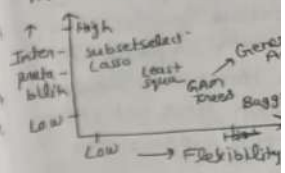
↳ If the chosen model is too far from the true f, then our estimate will be poor. We can use flexible models that can fit many diff. possible functional forms for f. But in general, fitting a more flexible model requires estimating greater no. of parameters. More complex models can lead to overfitting.

⇒ Non-Parametric methods do not make explicit assumption about functional form of f

↳ They have the potential to fit wider range of possible shapes for f

⇒ Less flexible = More restrictive because they can produce small range of shape to estimate f. Ex. Linear Regression.

→ If we are mainly ... much interpretable.

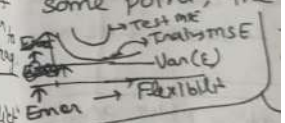


(*) $MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \ldots$

(*) We want to ch...

(*) The degrees of f... flexibility of a cu... fewer degrees of f...

(*) Training MSE ... Test MSE initially ... some point, the ...



(*) Overfitting hap... working too h... be caused by ... unknown funct^n f... very large because ... the training ...

(*) We always ...

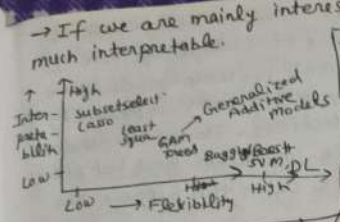
(*) Using CV (...

(*) U-shaped Tes... Test MSE = ...

(*) Variance refers ... $\hat{f}$ would change i... for f should not ... then small chang... methods have ...

(*) Bias refers ...

→ If we are mainly interested in inference then restrictive models are much interpretable.



→ Lasso relies on linear model but uses alternate fitting procedure & makes no. of coeff to exactly zero. So, Lasso is less flexible than Linear regressn & more interpretable than " " .

→ Trees, Bagging, Boosting, SVM, NNs are fully non-linear models.

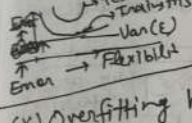
(*) $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$

(*) We are interested in the accuracy of the predictions that we obtain when we apply our method to previously unseen data.

(*) We want to choose the method that gives the lowest test MSE, not training MSE

(*) The degrees of freedom is a quantity that summarizes the flexibility of a curve. A more restricted & hence smoother curve has fewer degrees of freedom than a wiggly curve.

(*) Training MSE declines monotonically as flexibility increases. Test MSE initially declines as level of flexibility increases. However, at some point, the test MSE levels off & then starts to increase again.



→ U-shaped Test MSE is a fundamental property of statistical learning that holds regardless of particular dataset or method.

(*) Overfitting happens because statistical learning procedure is working too hard to find patterns in the training data that may be caused by random chance rather than by true properties of unknown functn f. When we overfit training data, test MSE will be very large because the supposed patterns that the method found in the training data simply don't exist in the test data.

(*) We always expect training MSE to be smaller than test MSE.

(*) Using CV (Cross-Validatn), we estimate ~~using~~ Test MSE using training data.

(*) U-shaped Test MSE is result of Bias-Variance Tradeoff.

Test MSE = Variance (+ve) + $Bias^2$ + Var(ε) = +ve. ⇒ We need low bias & low variance

↳ +ve & it is irreducible error.

(*) Variance refers to the amount by which $\hat{f}$ would change if we estimated it using diff. training data set. Ideally estimate for f should not vary too much b/w training sets. If a method has high variance then small changes in training data result in large changes in $\hat{f}$. More flexible methods have high variance..

(*) Bias refers to error introduced by approximating real life probm which may be complicated, by a simpler model. (Ex. real probm with linear model)

(*) As we use more flexible methods, variance will increase & bias will decrease. The relative rate of change of these 2 quantities determines whether the test MSE increases or decreases. Bias initially decreases faster than variance increases as flexibility increases & consequently test MSE declines. However, at some point increasing flexibility has little impact on bias but starts to significantly increase the variance. When this happens the test MSE increases.

(*) Theoretically, Bayes classifier produces the lowest possible test error rate, called Bayes error rate (analogous to irreducible error). But for real data, we don't know the conditional distribut^n of Y given X & so computing the Bayes classifier is impossible.

↳ KNN estimate the conditional dist^n of Y given X & then classify given observat^n to the class with highest estimated prob.

↳ KNN decision boundary is very close to that of Bayes classifier when k=1 → decision boundary is overly flexible. As k grows, the method becomes less flexible & produces approx linear decision boundary.

⇒ Choosing the correct level of flexibility is critical to success of ML models.

(*) Simple Linear Regression $Y \approx \beta_0 + \beta_1 X$ [Y = quantitative]

β_0: intercept, β_1: slope of linear model. (model coeff (parameters))

↳ we are regressing Y on X. Y is predicted on cal space (X)

From training data, we estimate $\hat{\beta}_0$ & $\hat{\beta}_1$ & predict using $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

→ There are many ways of measuring closeness but by far the most common approach involves minimizing the least squares criterion.

→ $e_i = y_i - \hat{y}_i$ & $RSS = e_1^2 + e_2^2 + \dots + e_n^2$. Using least square, minimize

$RSS = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \dots$ using calculus, we get

$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ where $\hat{\beta}_1 = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$

→ β_1 = slope = $\frac{\Delta y}{\Delta x}$ = the avg. increase in Y with 1 unit increase in X.

→ We typically assume error terms are independent of x.

If 'f' is to be approximated by a linear fun^n :-
$Y = \beta_0 + \beta_1 X + \epsilon$. (Population regression line)

→ The sample mean & population mean are different but the sample mean will provide a good estimate of population mean. If $E(\hat{\mu}) = \mu$ ⇒ Unbiased Estimator

↳ On one set $\hat{\mu}$ & μ might differ but avg on large no. of observat^ns gives μ.

→ Hence, unbiased estim
estimate the true
→ How far off will
given by standard e
→ $Var(\hat{\mu}) = SE(\hat{\mu}$
(This formula holds
→ Roughly, speaking,
estimat $\hat{\mu}$ differs
→ How close $\hat{\beta}_0$ & $\hat{\beta}_1$
$SE(\hat{\beta}_0)^2 = \sigma^2\left[\frac{1}{n} + \right.$
↳ These formula hol
common variance σ^2
→ The estimate of σ
$RSE = \sqrt{RSS/(n-2)}$.
→ SE. can be used
is defined as a rang
range will contain th
for β_1, 95% C.I. = $\hat{\beta}_1$
the assumption that
→ SEs can also be u
Ex. H_0: There is no rel
t-statistic ⇒ $t = \frac{\hat{\beta}}{S}$
~~small~~ → Compute prob.
absolute value, assum
→ Small p-value
⇒ Accuracy ⇒ (1) RSE
(Measure of fit)
↳ TSS measures total
TSS − RSS = measures a
explained (
↳ $R^2 \approx 1$ → Larger p

→ Hence, Unbiased estimator does not systematically over or under-estimate the true parameter.

→ How far off will the single estimate of $\hat{\mu}$ to μ. ⇒ Answer is given by standard error of $\hat{\mu}$ i.e. $SE(\hat{\mu})$

→ $Var(\hat{\mu}) = SE(\hat{\mu})^2 = \frac{\sigma^2}{n}$ [σ = std. dev. of each of the realizat^n y_i of Y]

[This formula holds provided that n observat^n are uncorrelated.]

→ Roughly, speaking, the std. error tells the avg. amount that this estimat $\hat{\mu}$ differs from actual value of μ.

→ How close $\hat{\beta}_0$ & $\hat{\beta}_1$ with true values β_0 & β_1.

$SE(\hat{\beta}_0)^2 = \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i-\bar{x})^2}\right]$ & $SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum(x_i-\bar{x})^2}$, $\sigma^2 = Var(\epsilon)$

↳ These formula holds when errors ϵ_i for each observat^n have common variance σ^2 and are uncorrelated.

→ The estimate of σ is known as RSE (Residual Std Error).

$RSE = \sqrt{RSS/(n-2)}$.

→ SE can be used to compute Confidence Intervals (C.I.). A 95% C.I. is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter. for β_1, 95% C.I. = $\hat{\beta}_1 \pm 2SE(\hat{\beta}_1)$. Similar for β_0. It is based on the assumption that errors are Gaussian.

→ SEs can also be used to perform Hypothesis tests, on the coeffs.

Ex. H_0: There is no relationship b/w X & Y. i.e. $\beta_1 = 0$ & $H_a: \beta_1 \neq 0$.

t-statistic ⇒ $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$ [t-dist^n has a bell-shape when $n > 30$ (normal dist^n)]

~~small~~ → Compute prob. of observing any no. equal to $|t|$ or larger in absolute value, assuming $\beta_1 = 0$. We call this prob. as p-value.

→ Small p-value = reject ~~Hbell~~ hyp. [p-value cutoff = 5% or 1%]

⇒ Accuracy ⇒ ① RSE ② R^2 statistic. (bcoz RSE is measured in units of Y)
(Measure of fit)

↳ $\frac{TSS - RSS}{TSS}$ [TSS = Total sum of square $= \sum(y_i - \bar{y})^2$]

↳ TSS measures total variance in Response Y. (amount of variability in response before performing regression)

TSS − RSS = measures amount of variability explained (removed).

↳ $R^2 \approx 1$ → Larger proportion of variability in response is explained. (R^2 is b/w 0 & 1)

→ if $R^2 \approx 0$ → Either model is wrong or error variance σ^2 is high.

→ Good R^2 value depends on appⁿ. → R^2 statistic is a measure of linear relationship b/w X & Y. Cor(X, Y) is also a " " " . So, For simple linear regressⁿ, $r = Cor(X, Y)$ & $R^2 = r^2$

⇒ Multiple Linear Regression = $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$

↳ β_J is the avg effect on Y of a one unit increase in X_J, holding all other predictors fixed.

↳ Null Hyp. H_0 : $\beta_1 = \beta_2 = \dots = \beta_p = 0$; H_a: atleast one β_J is non-zero.

↳ for this F-stat ≈ 1. O/w F-stat > 1 (relatⁿ exists) (no relatⁿ)

↳ F-statistic = $F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)}$

→ when $p > n$ ⇒ Forward selection can be used.

(*) Variable Selectⁿ: ① 2^p models (all subsets can be selected) & we can use various stats like Mallow's C_p, AIC, BIC, Adjusted R^2 to judge the quality of model. If p is large then 2^p is very big.

② Forward selectⁿ (Concept). Begin with Null model. then fit p simple linear regressⁿ & add to null model that has lowest RSS. & then fit 2 variable combinat from null & separate n-1 variables & then add variable with lowest RSS. Repeat until some stopping rule is satisfied.

③ Backward selection - Start with all variables in the model & remove variable with largest p-value (least significant variable) & repeat.

④ Mixed Selectⁿ = Combinatⁿ of forward & Backward [Generally used]

→ R^2 will always increase when more variables are added even those are weakly associated with resp. bcoz of always decrease in RSS. on training data.

(*) How much Y vary from $\hat{Y}$ → we use prediction intervals & it incorporate both reducible & irreducible error & wider than confidence interval.

(*) Qualitative Predictor = factor. has levels. → So, create dummy variable like (One-Hot Encoding). $X_i = 1$ &

(*) Additive assumption of Linear model means that the association b/w a predictor X_J & response Y does not depend on the values of other predictors. Linearity. assumption states that change in response Y associated with a 1 unit change in X_J is constant.

↳ Removing additive assumption ⇒ $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \underbrace{\beta_3 X_1 X_2}_{\text{Interactⁿ term}} + \varepsilon$.

→ The hierarchical principles state that if we include an interactⁿ in a model, we should also include the main effects even if p-values associated with their coeffs are not significant.

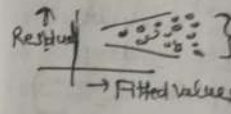
↳ $X_1 X_2$ is typically ca
tends to alter the
→ Non-Linear relationsh
But it is still a l

⇒ Potential Probⁿs

① Non-linearity of dat
Check whether pat

② Correlatⁿ of Erro
If there is correlat
tend to underestim
intervals will be na
Ex. if we double
would be same but
↳ correlatⁿ among
(plot residuals a

③ Non-constant va
The SE, C.I. & hyp
↳ We can check h
funnel shape in



④ Outliers → It
↳ occurs from i
↳ Identify using

⑤ High Leverage Poi
↳ leverage statistic

⑥ Collinearity = 2
Probⁿ. it can diffic
& it reduces acc
⇒ It is possible for
2 variables has high c
~~VIF = Robust~~.